

MoneyTranslate 2.0

Making financial letters understandable for people with low literacy

Hackathon 3 resit · SDG 10 Reduced Inequalities · Python + Google Gemini API · Milou

From feedback to MoneyTranslate 2.0

K1 – AI tool

"Plain API call, could be done in the web interface"

Gemini now reads photos, simplifies and structures every letter with a fixed, guarded prompt — and Python independently checks the amounts.

Problem

"Quite generic problem in all things money"

Narrowed to one situation: people with low literacy who receive payment reminders and debt letters with a deadline.

Completeness

"Could be more of a complete solution"

A real Streamlit app: paste text or upload a photo, Dutch/English, six fixed sections, verification and download.

Ethics

"Only mentions one specific issue"

Five prototype-specific risks, each with the consequence for the user, what I did, and the next step.

Half of people with debts struggle to read their letters

2.5 million
Dutch adults have difficulty with reading, writing and/or numeracy [1]

50.3%
of ~500 people with debts showed an indication of low literacy — against roughly 18% nationally [2][3]

747,500
households had problematic debts in 2024, and the number has risen in recent years [4]

Research among clients of Kredietbank Nederland shows that people with debts regularly do not understand the letters about their own financial problems — and only about 1 in 6 households with problematic debts is known to debt help [5].

Sources: [1] Stichting Lezen & Schrijven [2] Lezen ≠ Begrijpen, RUG/L&S 2018 [3] NOS 2018 [4] Tweede Kamer 2024-2025, nr. 3045 (CBS) [5] CBS 2018 — full list on last slide

The moment that matters: the 14-day letter

Day 0
A payment reminder arrives (the 'veertiendagenbrief')

→

Day 1–14
Last chance to pay the bill without extra costs

→

Day 15+
Collection costs are added: at least €40, up to 15% of the first €2,500

Concrete example: a €250 bill that is paid too late can legally cost €40 extra — 16% more [6]. Someone who cannot read the letter does not see the deadline or the warning, and the debt grows. **What goes wrong:** the information is in the letter, but the reader cannot get it out in time.

[6] Rijksoverheid, 'Hoogte incassokosten' — minimum €40, 15% over the first €2,500

Who MoneyTranslate is for — and who it is not for

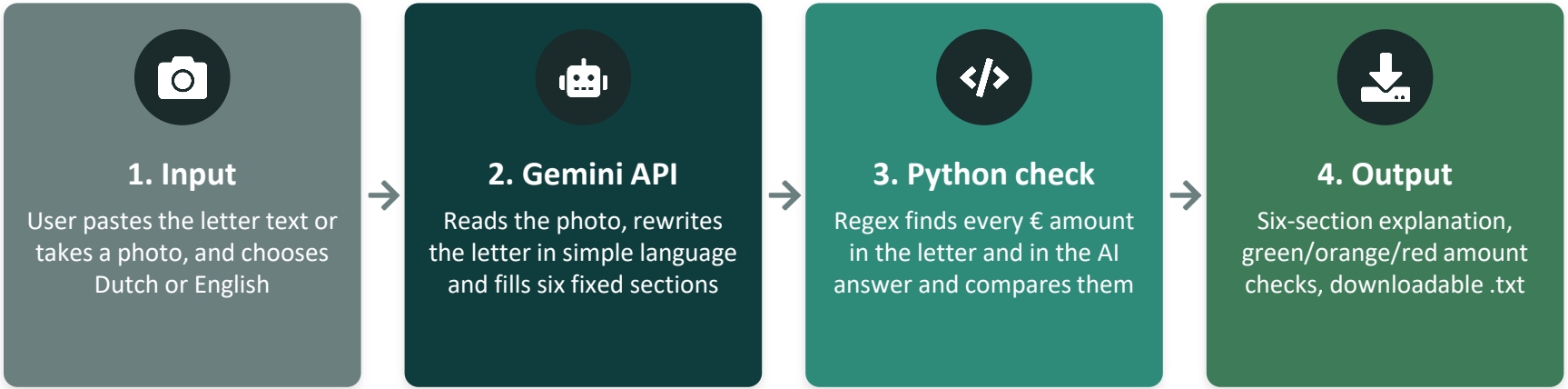
Intended users

- Adults in the Netherlands with low literacy (Dutch native speakers) or limited Dutch (second-language speakers)
- Receive letters from banks, municipalities or debt collectors with amounts and deadlines
- Can read short, simple sentences and use a smartphone camera
- Often feel ashamed to ask for help, so they leave letters unopened or misread them [5]

Not the intended user

- People who cannot read at all — they need audio (future step)
- People who need debt counselling or legal advice — the app refers to the original letter and professional help
- Companies or professionals handling business invoices
- Anyone without a phone, internet or basic digital skills

From letter to clear action list in four steps



The six fixed sections

1. Simple explanation	2. Important amounts	3. Deadlines	4. What should I do?	5. What is unclear?	6. Warnings / consequences

Why not just paste the letter into a chatbot?

Chatbot in a browser

- User must write a good prompt — hard for someone with low literacy
- Different answer format every time
- May guess unreadable parts or invent amounts
- No independent check: the user has to trust the AI
- Instructions hidden in a letter can steer the answer

MoneyTranslate

- No prompting needed: one button, one fixed prompt
- Always the same six sections, incl. a numbered action list
- Rules: never invent amounts, dates or consequences; say when a photo is unreadable
- Python compares every € amount in the letter with the AI answer
- Prompt tells Gemini to ignore instructions inside the letter

Without Gemini the app does not work: Python alone cannot read a photo, rewrite formal Dutch into simple language, or find which sentence contains the deadline and the warning.

Python checks the AI's amounts independently

```
# find € amounts in a text
amounts = re.findall(
    r"€\s?\d[\d.,]*", text)

# €1.250,00 and €1,250.00
# -> last separator = decimal
# -> both become 1250.0

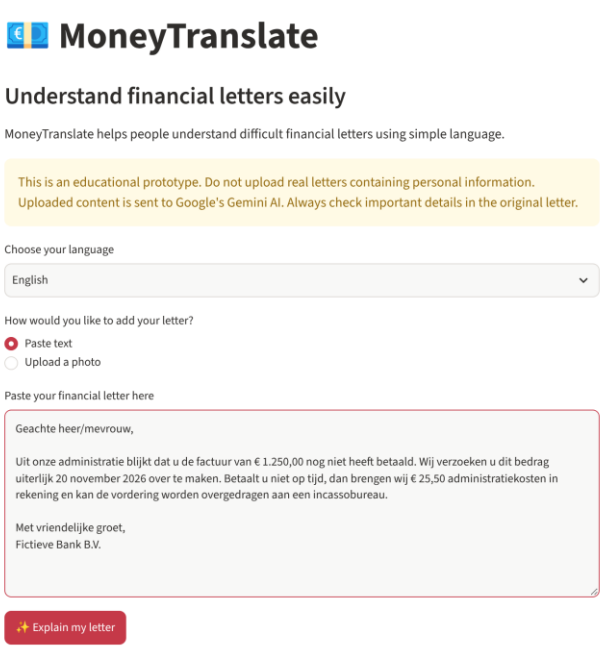
for amount in original_amounts:
    if amount in ai_amounts: OK
    else: WARN missing in answer
for amount in ai_amounts:
    if not in original: ERROR
```

Tested formats (real test run)

Input	Read as	
€1.250,00	1250.00	OK
€1,250.00	1250.00	OK
€ 25,50	25.50	OK
€25,-	25.00	OK
€1.250 (no cents)	1.25	Fix
25,50 euro / EUR 25,50	not found	Fix

Honest result: common formats work; two edge cases still need fixing (see next steps). It confirms amounts, not their meaning or deadlines.

Demo: a fictional payment reminder, end to end



1 — Input

Fictional bank letter: invoice €1.250,00, pay before 20 November 2026, otherwise €25,50 costs and a debt collector

2 — Gemini output

Six sections in simple English: what the letter is about, both amounts, the deadline, a numbered to-do list and the warning

3 — Python check

€1250.00 and €25.50 found in both texts — no amounts invented

4 — Photo route

Same letter as a photo: Gemini reads it; the app says amounts must be checked by hand

Screen recording of the full run: see the Hackathon 3 folder on GitHub. Run locally: python -m streamlit run app.py

A wrong answer for someone who cannot check it



The risk

Gemini gets a deadline or amount wrong. My users are exactly the people who cannot check the original letter — and the answer sounds confident.

Consequence for the user

They miss the 14-day deadline, pay too little or too late, and get at least €40 collection costs on top of the debt [6].

What I did

Prompt forbids inventing amounts, dates or consequences · section 'What is unclear?' · Python checks every amount · warning to always check the original letter.

What I would do next

Add the same check for dates · show the original sentence next to each amount · point to free help (e.g. the municipality's debt help) in every answer.

Four more risks specific to this prototype

Privacy

Letters contain names, IBANs and debts. Google's free API tier allows human review; Google says not to send personal data [7].

Done: warning in the app, only fictional letters used. Next: mask names/IBANs in Python before sending; paid tier.

Photos

A blurry photo can hide a deadline, and amounts from photos are not checked by Python.

Done: Gemini must flag unreadable text; user is told to check by hand. Next: transcribe first, then run the regex check.

Over-trust

Users may follow the action list without reading the letter or getting help.

Done: 'educational prototype, not financial or legal advice' warning. Next: link to free debt help in every answer.

New inequality

It needs a phone, internet and some digital skills; people who cannot read at all are not helped.

Next: read-aloud (text-to-speech) and testing with real users via a language café or 'Taalhuis'.

[7] Google, Gemini API Additional Terms of Service (Unpaid Services)

What I learned

A working API call is the start, not the product. Value came from a narrow target group, fixed structure, an independent check and honest limits.

Next steps

1. Fix the two parser edge cases (€1.250 without cents, 'euro'/'EUR')
2. Verify dates as well as amounts
3. Transcribe photos first so they can be verified too
4. Mask personal data before it is sent to Gemini
5. Read-aloud option and testing with real users

Sources

[1] Stichting Lezen & Schrijven (2019). Jaarverslag 2019 — 2.5 million adults have difficulty with reading, writing and/or numeracy.

[2] Keizer, M. (2018). Lezen ≠ Begrijpen: onderzoek naar leesvaardigheid onder mensen met financiële problemen. RUG / Stichting Lezen & Schrijven. Via gemeente.nu: 50.3% indication of low literacy.

[3] NOS (2018). 'Gormah kon niet lezen en kreeg zo een schuld van 50.000 euro' — national share ~18%.

[4] Tweede Kamer, Aanhangsel Handelingen 2024-2025, nr. 3045 — 747,500 households with problematic debts in 2024 (CBS).

[5] CBS (2018). Huishoudens met geregistreerde problematische schulden — ~1 in 6 known to debt help; Lezen ≠ Begrijpen focus groups.

[6] Rijksoverheid. 'Hoogte incassokosten' — minimum €40; €250 bill example.

[7] Google. Gemini API Additional Terms of Service — data use for Unpaid Services; ai.google.dev/gemini-api/terms.